



Panchip Microelectronics Co., Ltd.

PAN3029/3060 Series CAD Application Reference

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Documentation

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Revision History

Version	Revision date	Description
V1.0	2023.06	Initial release
V1.1	2023.08	Modify CAD Pins
V1.2	2024.03	Modify CAD function interface parameters
V1.3	2024.04	Modify function interface and chip description

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1 Function Introduction

PAN3029/3060 series chips support CAD-IRQ interrupts. After the CAD function is turned on and the Rx mode is entered, the chip will detect whether there is a ChirpIOT™ signal in the channel. If so, CAD-IRQ will be set high. The external MCU can determine whether there is a ChirpIOT™ signal in the channel by detecting whether the CAD-IRQ signal is pulled high within a certain period of time.

Users can read the CAD-IRQ signal through the GPIO port. The channel activity detection process is shown in Figure 1-1:

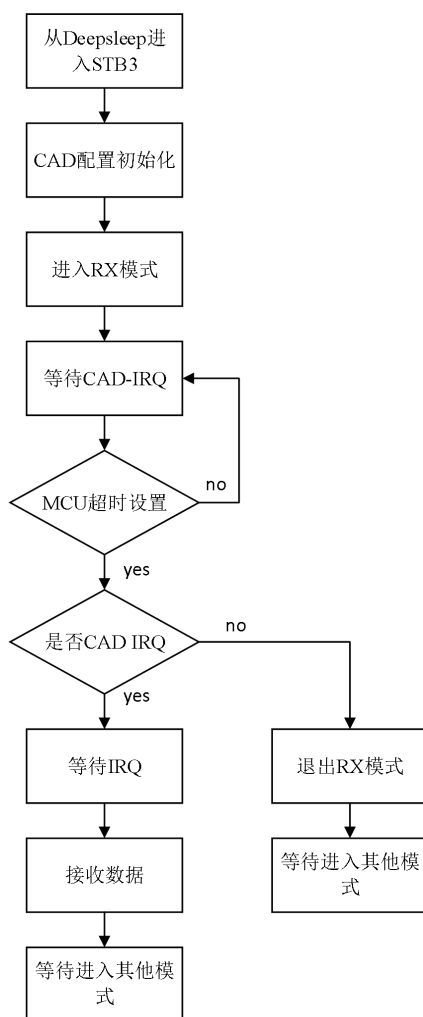


Figure 1-1 Channel Activity Detection (CAD)

2 Software Design Reference

2.1 Software Design Process

1. Chip initialization.
2. CAD initialization.
3. Enters receiving mode.
4. Observe the CAD-IRQ signal.

2.2 Software Design Verification

2.2.1 Verification steps

1. Sender module periodically sends data packets.
2. Receiver module is configured to receive mode.
3. Use an oscilloscope to capture the CAD-IRQ signal at the receiving side.

2.2.2 SDK Examples

Reference code:

```
ret = rf_init();
if(ret != OK)
{
    printf("  RF Init Fail");
    while(1);
}

rf_set_default_para();

rf_set_cad(CAD_DETECT_THRESHOLD_10, CAD_DETECT_NUMBER_3);
// Enable CAD mode and set Chirp-IoT signal threshold and number of valid preamble signals
rf_enter_continuous_rx();
```

```
while (1); //Wait for the oscilloscope to detect the CAD-IRQ signal
```

The sample code configures CAD initialization, configures GPIO11 as CAD detection IO, and then enters receiving mode.

The sender module periodically sends data packets (the duration of data packet preamble+payload is about 42ms), and uses a logic analyzer to capture the waveform of GPIO11 of the receiver module and observe the results of the logic analyzer.

2.2.3 Verification Results

The oscilloscope capture result is shown in Figure 2-1:

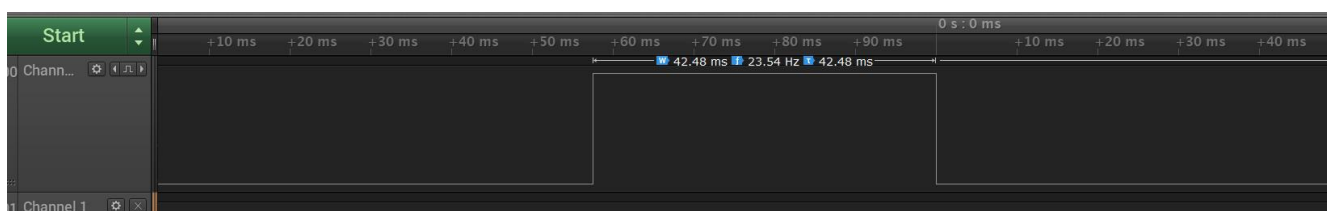


Figure 2-1 Oscilloscope capture result (CAD)

According to the results, when the sender module sends a data packet, the receiver module generates a CAD-IRQ, and the CAD detection pin GPIO11 is pulled high for about 42ms, which is the length of time for a complete ChirpIOT™ data packet.

3 Notes

3.1 How CAD affects the chip's receiving sensitivity

When the CAD function is initialized, the chip's receive signal detection threshold and CAD valid signal number threshold are modified. Setting different receive thresholds and valid signal numbers will affect the chip's receive sensitivity and may cause CAD false triggering.

```
uint32_t rf_cad_on(uint8_t threshold, uint8_t chirps)
{
    rf_set_gpio_output(11);
```

```
RF_ASSERT(rf_reset_spec_page_reg_bits(PAGE0_SEL, 0x5E, BIT6));

RF_ASSERT(rf_write_spec_page_reg_bits(PAGE1_SEL, 0x25, chirps, BIT0|BIT1));

RF_ASSERT(rf_write_spec_page_reg(PAGE1_SEL, 0x0f, threshold));

return OK;

}
```

To set the receiving threshold, you need to modify the parameters of the `rf_cad_on(uint8_t threshold, uint8_t chirps)` function. The effect of modifying the receiving threshold on the receiving sensitivity and false trigger probability is as follows (experimental data is tested in daily office application scenarios):

Table 3-1 Receive Threshold Settings

Receiving threshold	Number of valid signals	False trigger probability	Receiving sensitivity
0x0a	0x03	Low (once every few hours)	No impact
0x10	0x03	Very low (0 times in 24 hours)	Degradation 1dBm

When using the CAD function, users need to modify the parameters in the `rf_cad_on(uint8_t threshold, uint8_t chirps)` function according to the application scenario. After using the CAD function, it is recommended to call the `rf_cad_off()` function, which can turn off the CAD function and restore the receiving threshold.

3.2 About SDK and Demo System Board

The SDK provides the interface functions required for the CAD function. When CAD-IRQ is triggered, the detection pin GPIO11 will be pulled high. GPIO11 is connected to PA09 of the MCU on the demonstration system board.

```
#define GPIO_PIN_CAD    Pin09

#define GPIO_PORT_CAD   PortA

#define CHECK_CAD()     PORT_GetBit(GPIO_PORT_CAD, GPIO_PIN_CAD);
```

3.3 About the use of CAD

PAN3029/3060 can perform CAD detection on preamble and payload.

3.3.1 Preamble Check Method

When the complete preamble+payload signal arrives, the user can read the CAD-IRQ signal through the GPIO port at the receiving side. The CAD detection pin GPIO11 is pulled high for the duration of preamble+payload. At this time, the receiving side can produce correct rxdone results.

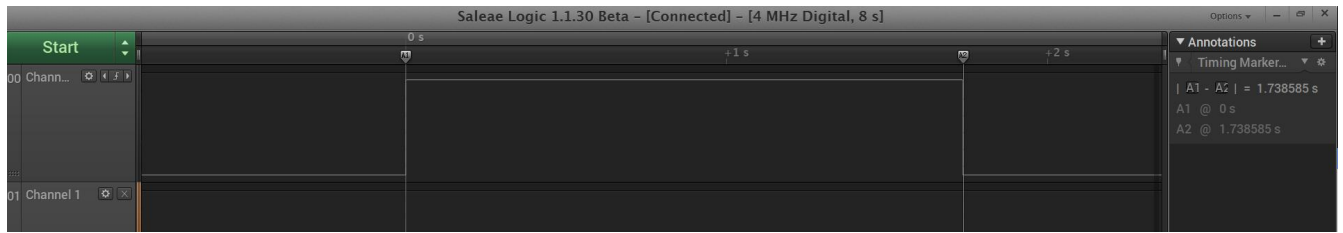


Figure 3-1 Logical capture result (complete preamble)

When an incomplete preamble+payload signal arrives (the transmitter transmits data first, and then the receiver turns on CAD detection during the preamble time period), the user can read the CAD-IRQ signal through the GPIO port at the receiver. There are two cases of changes in the CAD detection pin GPIO11:

1. If the preamble is relatively complete and the number of preambles contained inside is greater than the configured `chirp_num`, GPIO11 will be pulled high for the duration of preamble+payload (the preamble is incomplete, so the overall high time of GPIO11 is shorter than the situation in Figure 3-1). At this time, the receiver can generate a correct `rxdone` result.

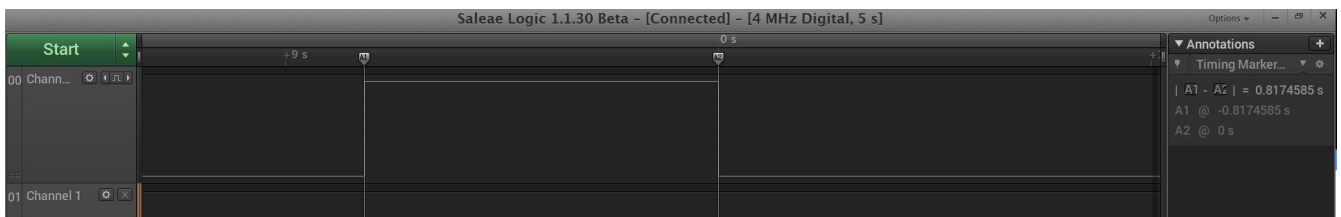


Figure 3-2 Logical capture result (more complete preamble)

2. If the preamble is incomplete and the number of preambles contained inside is less than the configured `chirp_num`, GPIO11 will always be at a low level. At this time, the receiving end will not produce a correct receiving result.
3. When there is only a payload signal, GPIO11 will always be at a low level. At this time, the receiving end will not produce a correct receiving result.

3.3.2 Software Application Reference

The duration of CAD detection is calculated in units of a single chirp duration, which is $2^{SF/BW}$ (SF

is the spreading factor, BW is the bandwidth in Hz, and time is in seconds).

Here are two software design references for typical CAD applications:

```
uint32_t rf_get_chirp_time(uint8_t bw, uint8_t sf)
{
    const uint32_t bw_table[4] = {62500, 125000, 250000, 500000};

    if(bw < BW_62_5K || bw > BW_500K)
    {
        return 0;
    }

    return (1000000 / bw_table[bw - BW_62_5K]) * (1 << sf);
}
```

1. The CAD function is used for channel detection before transmission to ensure that the current channel is idle, and then transmit data to avoid wireless signal collision interference and improve communication success rate.

CAD IO interrupt processing function sets the CAD event to ACTIVE.

```
void Cad_Irq_Callback(void)
{
    rf_set_cad_detect_flag(MAC_EVT_TX_CAD_ACTIVE);
}
```

The CAD timer timeout callback function sets the CAD event to TIMEOUT.

```
void cad_tx_timeout_cb(void)
{
    stimer_stop(&stimer_txcad_event);
    rf_set_cad_detect_flag(MAC_EVT_TX_CAD_TIMEOUT);
}
```

CAD detection function, configure to enter CAD receiving state, set CAD event to NONE, and start the timer.

```
uint32_t rf_cad_detect_start(void)
{
    uint32_t bw, sf;
    uint32_t one_chirp_time;
    bw = rf_get_bw();
    sf = rf_get_sf();
    one_chirp_time = rf_get_chirp_time(bw, sf); // us
    RF_ASSERT(rf_cad_on(CAD_DETECT_THRESHOLD_10,
CAD_DETECT_NUMBER_3));
    cad_tx_detect_flag = MAC_EVT_TX_CAD_NONE;
    RF_ASSERT(rf_enter_continuous_rx())
    SET_TIMER_MS(one_chirp_time * 7 / 1000 + 1);
    return OK;
}
```

Just wait and detect the CAD event. If it is TIMEOUT, you can transmit immediately. If it is ACTIVE, you need to retreat and wait for a while before transmitting. Before transmitting again, you need to detect the channel status again according to the above steps.

2. The CAD function is used for channel detection before receiving to check whether there is a useful signal in the current channel, and then decide to continue receiving, or turn off receiving and enter standby or sleep state to reduce power consumption.

```
bool check_cad_rx_inactive(uint32_t one_chirp_time)
{
    rf_delay_us(one_chirp_time * 7);
    rf_delay_us(360); // state machine start up time after enter rx state

    if (CHECK_CAD() != 1)
    {
        rf_set_mode(RF_MODE_STB3);
        return LEVEL_INACTIVE;
    }
}
```

```
    return LEVEL_ACTIVE;
}
```

Determine the CAD detection result according to the return value. If it is LEVEL_ACTIVE, continue receiving and wait for the reception result. If it is LEVEL_INACTIVE, you can turn off RX immediately to reduce power consumption.

```
//The main function can call check_cad_rx_inactive using the following logic
while (1)
{
    if(check_cad_rx_inactive(one_chirp_time) == LEVEL_ACTIVE)
    {
        //Signal detected, waiting for reception to complete
    }
    else{
        //No signal is detected, sleep can be performed
    }
}
```

For software application examples of check_cad_rx_inactive and rf_cad_detect_start, please refer to CAD routines.